

# Module 2: CSS Basics

## 1. Introduction to CSS

Cascading Style Sheets (CSS) is the language used to style the visual presentation of web pages. While HTML provides the structure, CSS allows you to control the layout, colors, fonts, and overall aesthetic.

Example:

```
h1 {
```

```
    color: #2d3436;
```

```
    text-align: center;
```

```
}
```

## 2. Types of CSS

There are three primary ways to apply CSS to an HTML document.

### Inline CSS

Applied directly within an HTML tag using the `style` attribute. It is generally used for unique styles on a single element.

Example:

```
<p style="color: blue; font-size: 14px;">This is an inline styled paragraph.</p>
```

### Internal CSS

Defined within a `<style>` element inside the `<head>` section of an HTML page. This is useful for styling a single document.

Example:

```
<head>
```

```
  <style>
```

```
    body { background-color: #f4f4f4; }
```

</style>

</head>

## External CSS

Linked via a `<link>` tag to a separate `.css` file. This is the preferred method for enterprise applications as it promotes reusability across multiple pages.

**Example:**`<link rel="stylesheet" href="styles.css">`

## 3. Selectors

Selectors are used to "find" the HTML elements you want to style.

| Selector Type    | Description                                                           | Example                                   |
|------------------|-----------------------------------------------------------------------|-------------------------------------------|
| Element Selector | Targets all elements of a specific type.                              | <code>p { color: red; }</code>            |
| Id Selector      | Targets a single unique element using the <code>#</code> symbol.      | <code>#header { padding: 10px; }</code>   |
| Class Selector   | Targets multiple elements with the same class using a <code>..</code> | <code>.btn { border-radius: 5px; }</code> |

## 4. Colors

CSS supports colors through names, Hex codes, RGB, and HSL values.

**Example:s**

`.text-primary { color: DodgerBlue; }`

`.text-hex { color: #4A90E2; }`

`.text-rgb { color: rgb(74, 144, 226); }`

## 5. Backgrounds

The background properties allow you to set colors or images as the backdrop for an element.

**\*\*Example:\*\***

```
```css
```

```
.hero-section {
```

```
  background-color: #ecf0f1;
```

```
  background-image: url('pattern.png');
```

```
  background-size: cover;
```

```
}
```

## 6. Fonts and Text Styling

Control the typography of your content to improve readability and branding.

**Example:s**

```
article {  
  font-family: 'Inter', sans-serif;  
  font-size: 16px;  
  line-height: 1.5;  
  text-decoration: none;  
  text-transform: capitalize;  
}
```

## ## 7. Margins and Padding

Spacing is critical for clean layouts.

\* **\*\*Padding:\*\*** Space inside the element (between content and border).

\* **\*\*Margin:\*\*** Space outside the element (between the border and other elements).

**\*\*Example:\*\***

```
```css
```

```
.card {
```

```
padding: 20px; /* Space inside */
```

```
margin: 15px; /* Space outside */
```

```
}
```

## 8. Borders

Borders define the edges of an element. You can specify width, style (solid, dashed, etc.), and color.

**Example:** `.input-field {`

```
border: 1px solid #dfe6e9;
```

```
border-radius: 4px;
```

```
}
```

## 9. CSS Box Model

The Box Model is the foundation of design and layout. Every HTML element is considered a box consisting of:

1. **Content:** The actual text or images.
2. **Padding:** The area around the content.
3. **Border:** The edge surrounding the padding.
4. **Margin:** The outermost layer, separating the box from neighbors.

**Example Visual Representation:** `.box {`

```
width: 300px;
```

```
padding: 10px;

border: 5px solid gray;

margin: 20px;

}

/* Total width = 300 + (10*2) + (5*2) = 330px */
```

## 10. Responsive Design Basics

Ensures web pages look good on all devices (desktops, tablets, and phones). This is primarily achieved through Media Queries.

**Example:**/\* Style for screens smaller than 600px \*/

```
@media screen and (max-width: 600px) {

    .sidebar {

        display: none;

    }

    .main-content {

        width: 100%;

    }

}
```

Congratulations! You have completed Module 2.